Document Number: WFH-LET-THE-AWI-FMR-26-005 – Part 1

Occupational Hygiene

Assessment work instruction / first mine report: – Thermal stress

30 June 2026

LETABA CRUSHERS (PTY) LTD
Farm Tarentaalrant 24LT
TZANEEN
Limpopo Province

Signature: _____
Appointed Occupational Hygienist
(Assessment instructor)

Signature: _____
Appointed Occupational Hygiene
Assistant (Assessor)

A Remedial action required from previous site visit

Date of previous site visit: -
24 to 26 June 2026.

Occupational Hygiene Assistant's observations / comments / non-conformances: -

- None.

B **Special / additional instruction/s by appointed Occupational Hygienist**

Date of instruction: -
30 June 2026.

- None.

1 Customer information

Date of field visit	Name of client	DME mine code
	Letaba Crushers (Pty) Ltd	14466

Mine Manager	E-Mail address	Mobile number

Site Occupational Hygienist	E-Mail address	Mobile number
Mr N B Letlape	ChiefOH@workforcehealthsa.co.za	+27 (0)83 456 2673

Occupational Medical Practitioner	E-Mail address	Mobile number

This report relates solely to the specific thermal stress monitoring identified herein. It does not imply WorkForce Health SA's approval of the conditions or performance of thermal stress monitoring other than the thermal stress monitoring referred to in this report, and the results do not apply to any similar thermal stress monitoring.

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2 Environmental conditions on the day of assessment

Start of shift: ___:___	
Temperature – Wet bulb	°C
Temperature – dry bulb	°C
Temperature - Globe	°C
WBGT Index	°C
Relative humidity	%
Wind speed	m/s
General Condition	

Middle of shift: ___:___	
Temperature – Wet bulb	°C
Temperature – dry bulb	°C
Temperature - Globe	°C
WBGT Index	C
Relative humidity	%
Wind speed	m/s
General Condition	

End of shift: ___:___	
Temperature – Wet bulb	°C
Temperature – dry bulb	°C
Temperature - Globe	°C
WBGT Index	°C
Relative humidity	%
Wind speed	m/s
General Condition	

Equipment used / serial number / comments		
WBGT monitor		WBGT Logger model AZ 87786
Anemometer		Kestrell pocket weather tracker model 4000

Note:
The weather conditions in the table above have been obtained from outside in the general environment.

3 Executive summary

Will only be submitted within ten working days from last day of site visit under "Part 2" of this report.

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5 Abbreviations / acronyms / definitions / terms

Term	Definition
LET	Letaba Crushers (Pty) Ltd.
MHSA	Mine Health and Safety Act, 29 of 1996 (as amended).
SANS	South African National Standards.
WBGT Index	Wet bulb globe temperature index – a composite heat stress index that combines wet bulb temperature, dry bulb temperature and globe temperature to assess environmental heat exposure risk.
Wet bulb temperature	The temperature measured using a thermometer with a moistened wick over its bulb (or calculated from the dry bulb temperature and humidity), reflecting the effect of evaporative cooling and used in assessing heat stress.
Dry bulb temperature	The ambient air temperature measured by a standard thermometer shielded from moisture and radiant heat.
Globe temperature	The temperature measured using a globe thermometer, reflecting radiant heat exposure in addition to ambient air temperature.
Anemometer	An instrument used to measure wind speed or air velocity.

6 Introduction

6.1 Purpose of this assessment: -

The purpose of thermal stress monitoring is to identify, assess, and manage the health and safety risks posed to workers by exposure to extreme heat or cold, ensuring compliance with the Mine Health and Safety Act, 29 of 1996 (as amended). It is a critical component of occupational hygiene and medical surveillance aimed at preventing heat-related illnesses (such as heat stroke and exhaustion) and maintaining worker well-being and productivity.

- Key purposes of thermal stress monitoring
  - Protecting worker health and safety: -
    - Monitoring ensures that employees are not exposed to hazardous thermal conditions, such as high ambient temperatures, high humidity, or intense solar radiation, which can cause fainting, heat stroke, or death.
  - Regulatory compliance: -
    - It is a legal requirement under the Mine Health and Safety Act, 29 of 1996 (as amended) for employers to conduct risk assessments and, where necessary, implement a mandatory code of practice for monitoring and managing thermal stress, specifically when conditions are deemed "abnormally hot".
  - Identifying hazardous "hot spots": -
    - Regular monitoring (e.g., using wet-bulb globe temperature or thermal work limit indices) helps identify areas on the surface that pose a high risk to personnel, particularly during summer.
  - Preventing productivity loss: -
    - High thermal stress leads to fatigue, poor concentration, and reduced performance. Monitoring helps in implementing work-rest cycles and adjusting schedules to maintain operational efficiency.
  - Guiding control measures: -
    - The data collected assists in determining the appropriate mix of engineering controls (e.g., shaded areas, ventilation), administrative controls (e.g., shift adjustments), and personal protective equipment.
  - Managing individual risk: -
    - It enables medical surveillance of workers, including heat tolerance screening and acclimatisation, to protect those more vulnerable due to age, health, or lack of training.

Thermal stress monitoring is not limited to underground mines, it is equally crucial for surface operations where workers are exposed to direct sunlight and high-intensity work.

6.2 Details of the assessor: -

Mr N B Ngobese

Practical certificate in mine environmental control with more than one year's experience.

Legal appointment as Occupational Hygienist Assistant at Letaba Crushers (Pty) Ltd.

6.3 Scope of this assessment: -

The scope of thermal stress monitoring on surface is regulated primarily under the Mine Health and Safety Act, 29 of 1996 (as amended), focusing on the identification, assessment, and management of heat and cold stress to protect worker health. While often associated with deep underground mining, thermal stress is a significant concern on surface due to environmental exposure, especially during hot summer months, requiring mandatory codes of practice.

- Key aspects of the scope
  - Mandatory regulatory compliance:
    - Regulation 9.2(7) of the Mine Health and Safety Act, 29 of 1996 (as amended) mandates that employers monitor and report thermal stress. This includes both hot and cold environments, requiring regular, often annual, monitoring, with increased focus on seasonal changes.
  - Identification of high-risk areas: -
    - Monitoring is not limited to, but specifically targets, areas with potential for high heat load, such as heavy machinery cabins, areas with high solar radiation, or near hot, broken rock in large-scale quarries and open-pit operations.
  - Environmental monitoring (heat stress): -
    - This involves measuring wet bulb globe temperature to account for air temperature, humidity, wind speed, and radiation. The goal is to monitor, in particular, the shift from "cool" to "hot" or "abnormally hot" conditions.
  - Cold stress management: -
    - Although often focused on heat, the scope includes monitoring for cold stress, particularly in specific regions or during winter months.
  - Worker health surveillance: -
    - The scope includes medical surveillance, particularly screening of workers for their ability to work in high-temperature environments.

- Implementation of controls: -
  - Monitoring data must feed into the mandatory code of practice, which includes implementing control measures such as providing adequate water, adjusting work-rest cycles, and ensuring proper personal protective equipment.

- Objectives of monitoring
  - To prevent heat-related illnesses (heat stroke, exhaustion) and cold-related injuries.
  - To ensure compliance with statutory reporting standards, which has seen increased rigor in recent years.

To manage risks arising from changing climatic conditions that affect outdoor workers.

7 Methodology

7.1 Methodology / strategy / monitoring procedure/s

The methodology for thermal stress monitoring involves a structured, risk-based approach outlined in the South African Mines Occupational Hygiene Codebook and guidelines from the Mine Health and Safety Council. The process aims to identify, evaluate, and control heat stress to protect workers' health through environmental, administrative, and personal protective measures.

- Risk assessment and classification

  Mines are required to implement a mandatory code of practice for thermal stress.
  - Subdivision: -
    - The mine is broken down into measurement areas, which are further divided into specific activity areas.
  - Assessment: -
    - Historical data on the thermal environment is used to categorise activity areas.
  - Risk categorisation: -
    - Each area is classified to define the level of risk, determining the required frequency of monitoring.

- Environmental monitoring methods

  Surface mining heat stress is largely driven by sun, machinery, and humidity.
  - Wet bulb globe temperature: -
    - This is the primary index used to assess and categorise thermal environments, measuring a combination of humidity, wind speed, and radiation.
  - Instrumentation: -
    - Handheld, calibrated equipment is used to measure dry bulb, wet bulb, and globe temperatures at specific locations on a regular schedule.
  - Real-time monitoring: -
    - For high-risk areas, continuous, automated monitoring stations are deployed to record data at frequent intervals, capturing daily and seasonal fluctuations.
  - Data analysis: -
    - Data is analysed to assess heat load and determine the need for control measures, with reports submitted as per category band.

- Physiological monitoring (worker health)
  As environmental monitoring does not account for individual variation, physiological monitoring is crucial.
  - Heat tolerance screening: -
    - Used to identify individuals at risk of heat illness.
  - Core temperature and heart rate monitoring: -
    - Technologies like ingestible sensors are used to monitor core body temperature directly.
  - Self-monitoring: -
    - Encouraging workers to track their own symptoms (e.g., thirst, fatigue, etc.).
- Administrative and engineering controls
  Monitoring results dictate the application of controls.
  - Work-rest schedules: -
    - Implemented based on wet-bulb-globe-temperature levels, requiring more frequent rest in high-heat conditions.
  - Engineering solutions: -
    - Including air-conditioned cabs for heavy machinery and hydration stations; and
    - Provision of shaded areas.
  - Acclimatisation: -
    - Procedures to safely prepare workers for hot conditions.
- Frequency of monitoring
  - Routine: -
    - Daily or regular checking of high-risk areas (e.g., near large, hot equipment or in deep pits).
  - Quarterly / annual reporting: -
    - Formal reporting of exposure levels to the Department of Mineral and Petroleum Resources.
  - Changes: -
    - Increased monitoring is required when new, significant heat sources (e.g., new machinery) are introduced.

The overall strategy emphasises that while monitoring is essential, controlling the heat at its source is the primary objective.

7.2 Equipment used: -

Equipment description	Calibration records	Limitations	Deviations

8 Results

8.1 Thermal stress – artificial (personal)

Date: ____ July 2026

Actual time		**Area / HEG number**	2.3	**HEG description**	Engineering maintenance – plant workshop (surface workshops)

Thermal stress - artificial heat (Personal)									
HEG description	Engineering maintenance (boiler making)								
Initials and surname of employee	**Identity number**	**Company number**	**Company occupation description and DMRE occupation code**	**Work and rest area**	**Time of measure-ment**	**Wet bulb °C** OEL 32.5° C	**Dry bulb °C** OEL 37.0° C	**Globe °C** OEL 37.0° C	**WBGT** OEL 34
				Work					
				Rest					
				Work					
				Rest					

Actual time		**Area / HEG number**	2.4	**HEG description**	Engineering maintenance – mechanical workshop (surface workshops)

Thermal stress - artificial heat (Personal)	
HEG description	Engineering maintenance (boiler making)

Initials and surname of employee	Identity number	Company number	Company occupation description and DMRE occupation code	Work and rest area	Time of measure-ment	Wet bulb °C OEL 32.5° C	Dry bulb °C OEL 37.0° C	Globe. °C OEL 37.0° C	WBGT OEL 34
				Work					
				Rest					
				Work					
				Rest					

Date: ____ July 2026

Actual time		Area / HEG number	2.3	HEG description	Engineering maintenance – plant workshop (surface workshops)

Thermal stress - artificial heat (Personal)									
HEG description	Engineering maintenance (boiler making)								
Initials and surname of employee	Identity number	Company number	Company occupation description and DMRE occupation code	Work and rest area	Time of measure-ment	Wet bulb °C OEL 32.5° C	Dry bulb °C OEL 37.0° C	Globe °C OEL 37.0° C	WBGT OEL 34
				Work					
				Rest					
				Work					
				Rest					

Actual time		Area / HEG number	2.4	HEG description	Engineering maintenance – mechanical workshop (surface workshops)

Thermal stress - artificial heat (Personal)									
HEG description	Engineering maintenance (boiler making)								
Initials and surname of employee	Identity number	Company number	Company occupation description and DMRE occupation code	Work and rest area	Time of measure-ment	Wet bulb °C OEL 32.5° C	Dry bulb °C OEL 37.0° C	Globe °C OEL 37.0° C	WBGT OEL 34
				Work					
				Rest					
				Work					
				Rest					

8.2 Thermal stress – ambient

Date: ____ July 2026

Work area

Warmest / coldest time during the shift

Actual time		Area / HEG number	1.1	HEG description	Opencast pit (drilling and blasting)

Parameter	Dry bulb temperature	OEL DB	Wet bulb temperature	OEL WB	Globe temperature	OEL GT	WBGT	OEL WBGT	Relative humidity	Wind speed	Wind chill	OEL wind

									%	m/s	°C	chill
Minimum												
Maximum		**37.0° C**		**32.5° C**		**37.0° C**		**34**				**4° C**
Average												

Rest area

Actual time: ____:____

Parameter	Dry bulb temperature	OEL DB	Wet bulb temperature	OEL WB	Globe temperature	OEL GT	WBGT	OEL WBGT	Relative humidity %	Wind speed m/s	Wind chill °C	OEL wind chill
Minimum												
Maximum		**37.0° C**		**32.5° C**		**37.0° C**		**34**				**4° C**
Average												

Work area

Warmest / coldest time during the shift

Actual time		**Area / HEG number**	2.1	**HEG description**	Crushing and screening (crushing)

Parameter	Dry bulb temperature	OEL DB	Wet bulb temperature	OEL WB	Globe temperature	OEL GT	WBGT	OEL WBGT	Relative humidity %	Wind speed m/s	Wind chill °C	OEL wind chill

Minimum		**37.0° C**		**32.5° C**		**37.0° C**		**34**				**4° C**
Maximum												
Average												

Rest area
Actual time: ____:____

Parameter	Dry bulb temperature	OEL DB	Wet bulb temperature	OEL WB	Globe temperature	OEL GT	WBGT	OEL WBGT	Relative humidity %	Wind speed m/s	Wind chill °C	OEL wind chill
Minimum		**37.0° C**		**32.5° C**		**37.0° C**		**34**				**4° C**
Maximum												
Average												

Work area
Warmest / coldest time during the shift

Actual time		Area / HEG number	2.2	HEG description	Sand plant (screening and grading)

Parameter	Dry bulb temperature	OEL DB	Wet bulb temperature	OEL WB	Globe temperature	OEL GT	WBGT	OEL WBGT	Relative humidity %	Wind speed m/s	Wind chill °C	OEL wind chill
Minimum												
Maximum		37.0° C		32.5° C		37.0° C		34				4° C
Average												

Rest area
Actual time: ____:____

Parameter	Dry bulb temperature	OEL DB	Wet bulb temperature	OEL WB	Globe temperature	OEL GT	WBGT	OEL WBGT	Relative humidity %	Wind speed m/s	Wind chill °C	OEL wind chill
Minimum												
Maximum		37.0° C		32.5° C		37.0° C		34				4° C
Average												

Work area
Warmest / coldest time during the shift

Actual time		Area / HEG number	2.3	HEG description	Engineering maintenance – plant workshop (surface workshops)

Parameter	Dry bulb temperature	OEL DB	Wet bulb temperature	OEL WB	Globe temperature	OEL GT	WBGT	OEL WBGT	Relative humidity %	Wind speed m/s	Wind chill °C	OEL wind chill
Minimum		37.0° C		32.5° C		37.0° C		34				4° C
Maximum												
Average												

Rest area
Actual time: ____:____

Parameter	Dry bulb temperature	OEL DB	Wet bulb temperature	OEL WB	Globe temperature	OEL GT	WBGT	OEL WBGT	Relative humidity %	Wind speed m/s	Wind chill °C	OEL wind chill
Minimum		37.0° C		32.5° C		37.0° C		34				4° C
Maximum												
Average												

Work area
Warmest / coldest time during the shift

Actual time		Area / HEG number	2.4	HEG description	Engineering maintenance – mechanical workshop (surface workshops)

Parameter	Dry bulb temperature	OEL DB	Wet bulb temperature	OEL WB	Globe temperature	OEL GT	WBGT	OEL WBGT	Relative humidity %	Wind speed m/s	Wind chill °C	OEL wind chill
Minimum												
Maximum		37.0° C		32.5° C		37.0° C		34				4° C
Average												

Rest area
Actual time: ____:____

Parameter	Dry bulb temperature	OEL DB	Wet bulb temperature	OEL WB	Globe temperature	OEL GT	WBGT	OEL WBGT	Relative humidity %	Wind speed m/s	Wind chill °C	OEL wind chill
Minimum												
Maximum		37.0° C		32.5° C		37.0° C		34				4° C
Average												

Work area
Warmest / coldest time during the shift

Actual time		Area / HEG number	2.5	HEG description	Management, administration, and support services

Parameter	Dry bulb temperature	OEL DB	Wet bulb temperature	OEL WB	Globe temperature	OEL GT	WBGT	OEL WBGT	Relative humidity %	Wind speed m/s	Wind chill °C	OEL wind chill
Minimum		37.0° C		32.5° C		37.0° C		34				4° C
Maximum												
Average												

Rest area
Actual time: ____:____

Parameter	Dry bulb temperature	OEL DB	Wet bulb temperature	OEL WB	Globe temperature	OEL GT	WBGT	OEL WBGT	Relative humidity %	Wind speed m/s	Wind chill °C	OEL wind chill
Minimum		37.0° C		32.5° C		37.0° C		34				4° C
Maximum												
Average												

Date: ____ July 2026
Work area
Warmest / coldest time during the shift

Actual time		Area / HEG number	1.1	HEG description	Opencast pit (drilling and blasting)

Parameter	Dry bulb temperature	OEL DB	Wet bulb temperature	OEL WB	Globe temperature	OEL GT	WBGT	OEL WBGT	Relative humidity %	Wind speed m/s	Wind chill °C	OEL wind chill
Minimum		37.0° C		32.5° C		37.0° C		34				4° C
Maximum												
Average												

Rest area
Actual time: ____:____

Parameter	Dry bulb temperature	OEL DB	Wet bulb temperature	OEL WB	Globe temperature	OEL GT	WBGT	OEL WBGT	Relative humidity %	Wind speed m/s	Wind chill °C	OEL wind chill
Minimum		37.0° C		32.5° C		37.0° C		34				4° C
Maximum												
Average												

Work area
Warmest / coldest time during the shift

Actual time		Area / HEG number	2.1	HEG description	Crushing and screening (crushing)

Parameter	Dry bulb temperature	OEL DB	Wet bulb temperature	OEL WB	Globe temperature	OEL GT	WBGT	OEL WBGT	Relative humidity %	Wind speed m/s	Wind chill °C	OEL wind chill
Minimum												
Maximum		37.0° C		32.5° C		37.0° C		34				4° C
Average												

Rest area
Actual time: ____:____

Parameter	Dry bulb temperature	OEL DB	Wet bulb temperature	OEL WB	Globe temperature	OEL GT	WBGT	OEL WBGT	Relative humidity %	Wind speed m/s	Wind chill °C	OEL wind chill
Minimum												
Maximum		37.0° C		32.5° C		37.0° C		34				4° C
Average												

Work area
Warmest / coldest time during the shift

Actual time		Area / HEG number	2.2	HEG description	Sand plant (screening and grading)

Parameter	Dry bulb temperature	OEL DB	Wet bulb temperature	OEL WB	Globe temperature	OEL GT	WBGT	OEL WBGT	Relative humidity %	Wind speed m/s	Wind chill °C	OEL wind chill
Minimum												
Maximum		37.0° C		32.5° C		37.0° C		34				4° C
Average												

Rest area
Actual time: ____:____

Parameter	Dry bulb temperature	OEL DB	Wet bulb temperature	OEL WB	Globe temperature	OEL GT	WBGT	OEL WBGT	Relative humidity %	Wind speed m/s	Wind chill °C	OEL wind chill
Minimum												
Maximum		37.0° C		32.5° C		37.0° C		34				4° C
Average												

Work area
Warmest / coldest time during the shift

Actual time		Area / HEG number	2.3	HEG description	Engineering maintenance – plant workshop (surface workshops)

Parameter	Dry bulb temperature	OEL DB	Wet bulb temperature	OEL WB	Globe temperature	OEL GT	WBGT	OEL WBGT	Relative humidity %	Wind speed m/s	Wind chill °C	OEL wind chill
Minimum												
Maximum		37.0° C		32.5° C		37.0° C		34				4° C
Average												

Rest area
Actual time: ____:____

Parameter	Dry bulb temperature	OEL DB	Wet bulb temperature	OEL WB	Globe temperature	OEL GT	WBGT	OEL WBGT	Relative humidity %	Wind speed m/s	Wind chill °C	OEL wind chill
Minimum												
Maximum		37.0° C		32.5° C		37.0° C		34				4° C
Average												

Work area
Warmest / coldest time during the shift

Actual time		Area / HEG number	2.4	HEG description	Engineering maintenance – mechanical workshop (surface workshops)

Parameter	Dry bulb temperature	OEL DB	Wet bulb temperature	OEL WB	Globe temperature	OEL GT	WBGT	OEL WBGT	Relative humidity %	Wind speed m/s	Wind chill °C	OEL wind chill
Minimum		37.0° C		32.5° C		37.0° C		34				4° C
Maximum												
Average												

Rest area
Actual time: ____:____

Parameter	Dry bulb temperature	OEL DB	Wet bulb temperature	OEL WB	Globe temperature	OEL GT	WBGT	OEL WBGT	Relative humidity %	Wind speed m/s	Wind chill °C	OEL wind chill
Minimum		37.0° C		32.5° C		37.0° C		34				4° C
Maximum												
Average												

Work area
Warmest / coldest time during the shift

Actual time		Area / HEG number	2.5	HEG description	Management, administration, and support services

Parameter	Dry bulb temperature	OEL DB	Wet bulb temperature	OEL WB	Globe temperature	OEL GT	WBGT	OEL WBGT	Relative humidity %	Wind speed m/s	Wind chill °C	OEL wind chill
Minimum		37.0° C		32.5° C		37.0° C		34				4° C
Maximum												
Average												

Rest area
Actual time: ____:____

Parameter	Dry bulb temperature	OEL DB	Wet bulb temperature	OEL WB	Globe temperature	OEL GT	WBGT	OEL WBGT	Relative humidity %	Wind speed m/s	Wind chill °C	OEL wind chill
Minimum		37.0° C		32.5° C		37.0° C		34				4° C
Maximum												
Average												

9 Field Officer's observations / comments

10 Acknowledgement

10.1 We acknowledge that the above assessment/s have been conducted in the relevant areas and that all information, e.g., descriptions and references to work areas / equipment contained in this document are correct.

10.2 We noted / agreed upon certain remedial action/s required, as listed in the table below, and that the relevant target dates provided will be met.

10.3 We further acknowledge that once remedial action is completed, we will inform the appointed Occupational Hygienist, via email, administrator@workforcehealthsa.co.za, to ensure re-assessment / re-evaluation is done during the next site visit.

Responsible person	Detail of remedial action required	Target date	Responsible person signature and date of completion

11 Acknowledgement and confirmation of receipt

_____ **Mine Manager**	_____ **Engineer / Competent Person**
_____ **Employee Representative**	_____ **SHE Representative**
_____	_____
_____ **WFH Occupational Hygiene Assistant**	_____ **Date**